Reg. No. | | | | | | | | | | | [13EC61/12EC61]

SIXTH SEMESTER B.Tech/B.E UNIVERSITY EXAMINATION, JUNE – 2016

DIGITAL COMMUNICATION

Time: 3 Hours **Maximum Marks:70**

Instructions: Answer any FIVE full questions

1. a. Discuss the advantages and disadvantages of Digital communication system **06**

b. State and prove sampling theorem with necessary equations. **08**

2. a. How low pass signal will be reconstructed from the knowledge of samples? **10**

b. A low pass signal x(t) has a spectrum x(f) given by **04**

$$x(t)=\begin{cases}1-\dfrac{|f|}{200}, & |f|<200\\ 0, & \text{elsewhere}\end{cases}$$

i. Assume that x(f) is ideally sampled at f_x=300Hz, sketch the spectrum of $x_s(t)$

ii. Repeat part(i) with with f_s= 400Hz

3. a. With neat diagram explain the Delta modulation techniques and discuss the errors in the Delta modulation **02**

b. Prove that signal to quantizing noise ratio for sinusoidal modulation is $(SNR)_{dB}=1.8+6n$ **08**

c. Explain Delta Modulation with staircase approximation. **04**

4. a. Explain inter symbol interference with necessary equations. **06**

b. Explain the modes of operation of an adaptive equalizer. **08**

5. a. Explain signal space diagram for coherent binary PSK system. **06**

b. Derive an expression for the average probability of symbol error for coherent binary FSK system. **08**

6. a. Explain differential phase – shift keying with necessary diagram and waveform. **08**

b. Set up a block diagram for the generation of a binary FSK signal S(+) with continuous phase by using the representation. **06**

$$S(t)=\sqrt{\frac{2E_b}{T_b}}\cos\left(\frac{\pi t}{T_b}\right)\cos(2\pi f_c t)\mp\sqrt{\frac{2E_b}{T_b}}\sin\left(\frac{\pi t}{T_b}\right)\sin(2\pi f_c t)$$

7.	**a.**	Explain correlation receiver with necessary diagram.	**07**
	b.	Explain the two properties of matched filters.	**07**
8.	**a.**	For a three stage feedback shift register having initial state of shift register 100 then, find the PN sequence code and verify maximal length sequence properties for the generated PN sequence.	**08**
	b.	Distinguish between fast hopping and slow hopping technique in spread spectrum modulation.	**06**